

# Learning from Zhong-Can Ou-Yang's work on red blood cells

Curvature elasticity, biconcavity, and stability

These study notes follow the route from a membrane energy to a red-cell-like shape. The central lesson is methodological: derive force balance, enforce the geometric constraints, and then test stability. An analytic contour is valuable, but its resemblance to a cell does not settle all three questions.

## 1 Start with the membrane energy

Use the notation of the membrane-fluctuation note:  $F$ ,  $\kappa$ ,  $\bar{\kappa}$ ,  $H$ ,  $K$ ,  $C_0$ , and  $\sigma$ . For a closed membrane, include the volume constraint as well:

$$F = \int dA \left[ \frac{\kappa}{2} (2H - C_0)^2 + \bar{\kappa} K + \sigma \right] + \Delta p \int dV. \quad (1)$$

Here the normal points outward,  $H$  is positive on an outward-oriented sphere, and  $\Delta p$  is exterior minus interior pressure. The spontaneous-curvature sign follows the displayed energy. The multipliers enforce prescribed area and volume; their values must be found with the shape. Spatially constant moduli and a fixed topology are assumed. For a smooth closed surface of spherical topology,  $\int K dA = 4\pi$ , so the Gaussian term does not affect local force balance.

Ou-Yang and Helfrich derived the general equilibrium equation and the higher variations needed for deformation and stability studies [1]. In these conventions, normal force balance reads

$$\boxed{\Delta p + 2\sigma H - \kappa(2H - C_0)(2H^2 - 2K + C_0H) - 2\bar{\kappa}\nabla^2 H = 0.} \quad (2)$$

The Laplacian here acts on the curved membrane, rather than on its projection onto a plane. Curvature already contains second derivatives of the shape, so the last term makes this a fourth-order shape equation. Changing the curvature convention changes several signs together, not just the sign of one term.

## 2 A quick consistency check

For a sphere with  $C_0 = 0$ , curvature is constant and  $H^2 = K$ . Both bending terms in Eq. (2) vanish, leaving

$$\Delta p + 2\sigma H = 0. \quad (3)$$

Thus a positive tension requires a greater interior pressure, as expected from Laplace's law. The bending contribution itself is  $8\pi\kappa$ , independent of sphere size. This is a useful check on signs and factors of two, not a prediction that a red cell must be spherical. At fixed area, a sphere encloses the largest volume; a deflated membrane has room to adopt nonspherical shapes.

### 3 What the biconcave analytic solution teaches

Naito, Okuda, and Ou-Yang used a rigorous axisymmetric solution to distinguish between competing reductions of the same shape problem [2]. The lesson is stronger than obtaining an attractive contour: a proposed reduction must preserve the admissible solutions of the original variational equation.

The construction describes the meridian through its tangent angle. In the analytic family, its sine contains a term proportional to radial distance times a logarithm of radial distance relative to a fixed scale. Integrating the corresponding slope gives the profile; revolving it gives the surface. A suitable branch has a depressed center and a thick rim. The logarithm is part of the solution, not a polynomial fit to a micrograph.

In the special biconcave construction,  $\Delta p = 0$  and  $\sigma = 0$ , while spontaneous curvature is nonzero. These are conditions on a particular solution family, not a general statement that physiological red cells have zero tension or pressure difference. Liu and collaborators, including Ou-Yang, explored its broader family of profiles and energy comparisons [3]. A branch must also close properly and avoid self-intersection before it can represent a cell membrane.

**Two qualifications matter.** First, an energy comparison within one analytic family is not a proof of the global minimum at a specified area and volume. Second, this familiar logarithmic construction has singular curvature at the two poles. A profile can look smooth at plotting resolution while failing the regularity required of a classical closed Helfrich equilibrium. Modern analysis explicitly distinguishes these biconcave discoids from smooth Helfrich surfaces [4]. The exact solution is therefore an instructive benchmark with a pole qualification; it is not a complete smooth, force-free model of a living red cell.

### 4 Stationarity is only the first test

The shape equation expresses  $\delta F = 0$ . Local stability additionally requires  $\delta^2 F \geq 0$  for admissible perturbations, with the constraints enforced to the appropriate order. A negative direction in the second variation indicates a deformation that lowers the energy. Testing only axisymmetric perturbations can miss a symmetry-breaking instability.

This distinction is explicit in the work of Naito, Okuda, and Ou-Yang on pressure-induced polygonal deformations [5]. They evaluated the second variation about biconcave vesicles and analyzed transitions to elliptical and higher polygonal shapes. The comparison concerned observed liposome transformations. It should not be retold as a direct measurement of an identical instability threshold in human red cells.

## 5 Connecting the theory to real red cells

The fluid-bilayer model supplies curvature elasticity, but a red-cell membrane also includes a protein skeleton with shear and stretching elasticity. Lim, Wortis, and Mukhopadhyay showed how combining those contributions with bilayer mechanics can reproduce a sequence from cup-like to biconcave to spiculated shapes [6]. This explains why a successful fluid-vesicle calculation is a starting point rather than a full account of cell mechanics.

For a quantitative comparison, specify the enclosed volume, membrane area, leaflet asymmetry, and external loading. State whether the skeleton is included. A static energy does not determine the speed of deformation: that also requires dissipation and transport. These distinctions connect this topic to the fluctuation and osmosis notes without treating an equilibrium calculation as a dynamical theory.

## 6 A practical reading and calculation sequence

1. Read Ou-Yang and Helfrich for the separation between the first and second variations. Check the pressure and curvature signs against the sphere before comparing formulas.
2. Read Naito, Okuda, and Ou-Yang's counterexample as a test of mathematical equivalence. Track what happens to integration constants and pole conditions when reducing the equation.
3. Follow the analytic family in Liu and collaborators. Separate the existence of a biconcave branch from admissibility, constrained energy comparisons, and stability.
4. Read the polygonal-deformation study to see why perturbations beyond rotational symmetry matter. Then compare with a model that includes the membrane skeleton.

**Checks to keep beside a calculation:** Does the surface close? Are both poles regular? Are area and volume the intended ones? Does substitution satisfy the shape equation? Which perturbations were tested? These questions are more informative than visual agreement alone.

## References

- [1] Z.-C. Ou-Yang and W. Helfrich, "Bending energy of vesicle membranes: General expressions for the first, second, and third variation of the shape energy and applications to spheres and cylinders," *Phys. Rev. A* **39**, 5280–5288 (1989). doi:10.1103/PhysRevA.39.5280.
- [2] H. Naito, M. Okuda, and Z.-C. Ou-Yang, "Counterexample to some shape equations for axisymmetric vesicles," *Phys. Rev. E* **48**, 2304–2307 (1993). doi:10.1103/PhysRevE.48.2304.
- [3] Q.-H. Liu, H.-J. Zhou, J.-X. Liu, and Z.-C. Ou-Yang, "Spheres and prolate and oblate ellipsoids from an analytical solution of the spontaneous-curvature fluid-membrane model" (1999). arXiv:cond-mat/9906038.
- [4] R. López, B. Palmer, and A. Pámpano, "Axially symmetric Helfrich spheres," *Commun. Pure Appl. Anal.* **29**, 163–192 (2026). doi:10.3934/cpaa.2026021.
- [5] H. Naito, M. Okuda, and Z.-C. Ou-Yang, "Polygonal shape transformation of a circular biconcave vesicle induced by osmotic pressure," *Phys. Rev. E* **54**, 2816–2826 (1996). doi:10.1103/PhysRevE.54.2816.
- [6] G. H. W. Lim, M. Wortis, and R. Mukhopadhyay, "Stomatocyte-discocyte-echinocyte sequence of the human red blood cell: Evidence for the bilayer-couple hypothesis from membrane mechanics," *PNAS* **99**, 16766–16769 (2002). doi:10.1073/pnas.202617299.